FYP MEETING LOG

Industry Meeting - Inspired Square (Kick-off)

Agent-Based Game Analysis in Unity Based Upon Curiosity and RND Exploration Using PPO and SAC
Student: Raza Ullah Khan (NUM-BSCS23-28) | Supervisor: Dr. Malik M Ali Shahid | Co-Supervisor: Mr. Abdul Rafay | Industrial Partner: Inspired Square

Meeting No.	01 (Industry Meeting 1 of 1)
Meeting With	Industrial Partner - Inspired Square
Date	Wednesday, 7 October 2026
Time	10:00 AM - 11:00 AM
Duration	1 hour (60 minutes)
Mode / Venue	Online video call (Google Meet)
Attendees	Raza Ullah Khan (Student); Representative, Inspired Square (Game Development Team)
Milestone	FN-01 (Month 1, Weeks 1-2): Project Initialization, Development Environment & Version Control

1. Purpose of the Meeting

This was our first meeting with the industrial partner. I wanted to explain the project in simple words, hear what the studio expects from it, and agree on how we will keep them updated during the eight months.

2. Agenda

1. Short introduction of the project and the research question.
2. Why Hummingbirds is the first environment and what the agent has to do.
3. What the studio finds useful from this kind of work.
4. Tools, training speed and practical tips from the studio side.
5. How and when we share progress, and what is out of scope.

3. What We Discussed

- **Project in one line.** I explained that we are not making a new game. We are training agents in Unity with two learning methods (PPO and SAC) and three ways of exploring (no bonus, Curiosity, and RND), so six combinations in total, and then we compare them fairly.
- **Why it matters to a studio.** The representative said studios are interested in agents that can play a level by themselves, because such agents can act like automatic play-testers. An agent that explores well can find places in a level that a scripted bot never visits. This is the main reason exploration (Curiosity and RND) is interesting for them.
- **Hummingbirds as the first game.** The bird has to fly in a 3D area and drink nectar from flowers. It is small, well known in the ML-Agents community and quick to train, so it is a safe first step before a second, harder game.
- **Watch for reward hacking.** They warned that agents often find strange shortcuts to get reward. Their advice: do not only look at graphs, also record short videos of the trained agent and actually watch how it behaves.
- **Training speed tips.** Train with a built game (not inside the Editor), switch off graphics during training, and run several copies of the environment in parallel. This saves a lot of time when we reach 30 runs.
- **Keep it neutral.** They liked that the proposal does not assume a winner. They asked us to report honestly, even if the extra exploration does not help in some cases.
- **Scope and data.** No commercial game or private studio data will be used in this project. Using the findings on the studio's own games is only future scope.

4. Small Implementation Part

To make the discussion concrete, I shared the planned experiment matrix and the naming rule for training runs. I also showed the type of command we plan to use once the Hummingbirds build is ready.

Algorithm	Baseline (no bonus)	Curiosity (ICM)	RND
PPO	Condition 1	Condition 2	Condition 3
SAC	Condition 4	Condition 5	Condition 6

Table: six experiment conditions shared with the studio. Every condition will be repeated with several random seeds.

```
# Planned run naming rule:  <algorithm>_<exploration>_s<seed>
#   e.g. ppo_base_s1, ppo_curiosity_s1, sac_rnd_s3

# Planned training command (built game, no graphics, 4 copies in parallel)
mlagents-learn config/trainer_config.yaml --run-id=ppo_base_s1 --seed=1 ^
    --env=Builds/Hummingbird --num-envs=4 --no-graphics
```

5. Decisions Taken

- Hummingbirds stays as the main environment for the full six-condition comparison.
- Every trained agent will also be checked visually with a short recorded video, not just with graphs.
- Training will be done on a built game with graphics turned off and parallel environments where the PC allows it.
- A short progress update will be sent to the studio once a month, with a demo video when there is something to show.
- Studio games and private data are out of scope; any real use stays as future work.

6. Action Items

#	Action Item	Responsible	Due Date
1	Share the updated proposal and the six-condition table with the studio.	Raza	9 Oct 2026
2	Brief the supervisor about the studio's suggestions.	Raza	12 Oct 2026
3	Add 'watch agent videos' and 'headless training' to the experiment plan.	Raza	16 Oct 2026
4	Suggest any small game ideas that could serve as the second environment.	Inspired Square	Before Month 5

7. Other Notes

- The representative offered to review the final results before the defence (planned at milestone FN-15).
- Contact for monthly updates will be by email.

8. Next Meeting

Next industry update: monthly email in the first week of November 2026. Next full meeting: when baseline results are ready (around Month 3).